Machine Learning 2025

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1 What is Machine Learning?

1.1 What is Machine Learning?

Machine learning aims at programming a computer to learn from information (Data) to perform a task (Prediction, etc.) in an optimal way in terms of a performance measure (Loss Function).

Recent success of machine learning due to advances in its main ingredients:

1. Data availability: huge growth due to automation, digitalization, web applications.
2. Computing power: significant increase thanks to GPU computing, parallelization, etc.
3. The learning algorithms: theoretical developments of new statistical models and training methods.

1.2 The statistical framework

Types of variables:

- quantitative variables take values in an ordered set, typically the real numbers R;
- qualitative variables (also called categorical or factor) take values in a finite set $G = G1, ..., Gq$ without ordering, e.g., $Yes, No, blue, red, green, yellow$, etc.

Supervised learning

We observe training data $(x_i, y_i), i = 1, ..., n$ where

- the inputs $xi \in Rp$ are called predictors, covariates or features;
- the outputs y_i are called responses (in this course we assume they are univariate).

The goal is to predict the (unknown) response y_0 for a new (known) predictor x_0.

Regression: the responses $y_i \in R$ are quantitative.

Classification: the responses $y_i \in G$ are qualitative.

Unsupervised learning

We only observe multivariate data $x_i \in R^p, i = 1, ..., n$ but without a particular response.

The goal is to summarize, understand, interpret and visualize the data.

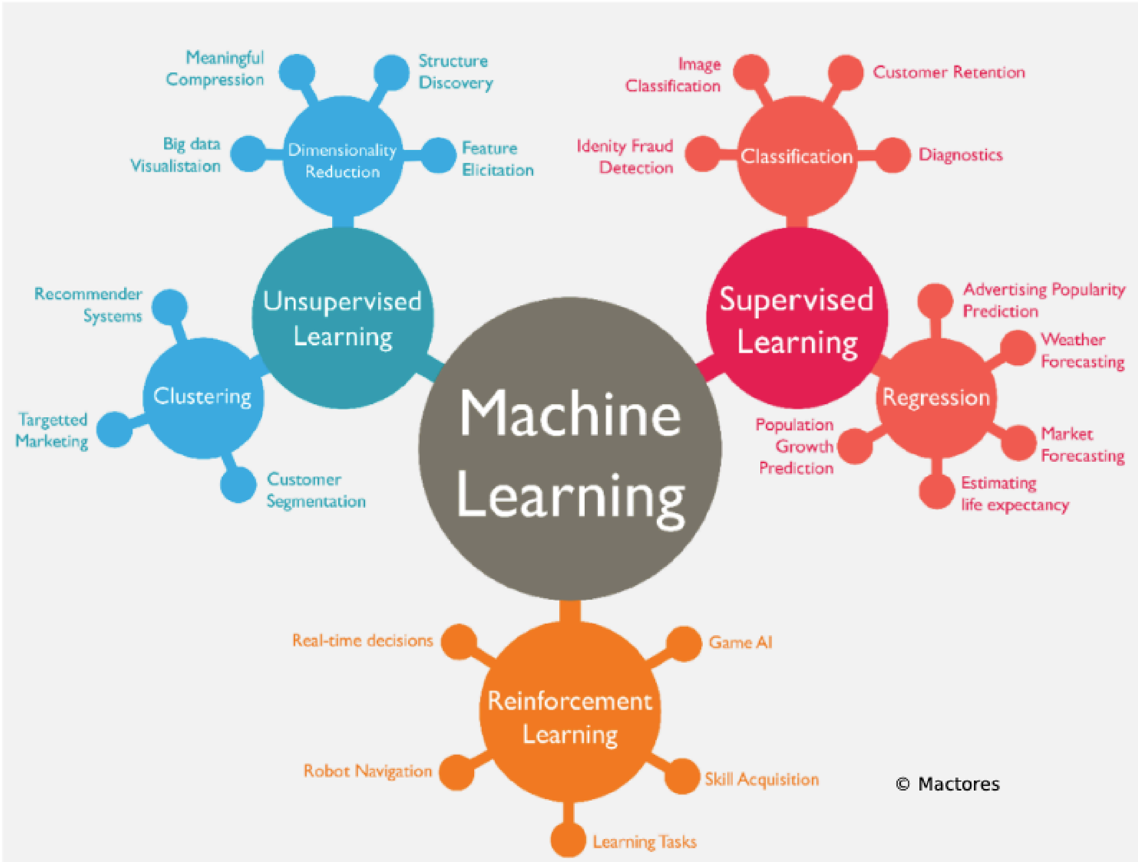


Figure 1: Machine Learning: an overview

1.3 Regression: simple linear model

- **Data**: Observations (x1, y1), . . . , (xn, yn)
- **Goal**: Predict unknown outcome y0 for new predictor x0.

1.4 Linear regression

$$\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$$

- $\hat{y}_0$: This is the predicted value of the response variable y when the input is x_0. The "hat" (^) indicates it's an estimate, not the true value.

- $\widehat{\beta}_0$: This is the estimated intercept.
  It represents the predicted value of y when $x = 0$. It's called an *estimate* because it's computed from data and is an approximation of the true underlying intercept β_0.
- $\widehat{\beta}_1$: This is the estimated slope (also an approximation). It tells us how much y is expected to increase (or decrease) when x increases by 1 unit.
- x_0: This is the value of the predictor (input) variable that we're using to make a prediction.

The numbers $\widehat{\beta}_0, \widehat{\beta}_1 \in \mathbb{R}$ are estimated model parameters. ### What does $\widehat{\beta}_0, \widehat{\beta}_1 \in \mathbb{R}$ mean?

It says both $\widehat{\beta}_0$ and $\widehat{\beta}_1$ are **real numbers** — elements of the set of real numbers $\mathbb{R}$.

In practical terms: they're just numerical values (like 2.3, -1.5, etc.) estimated from the data using statistical methods (usually least squares).

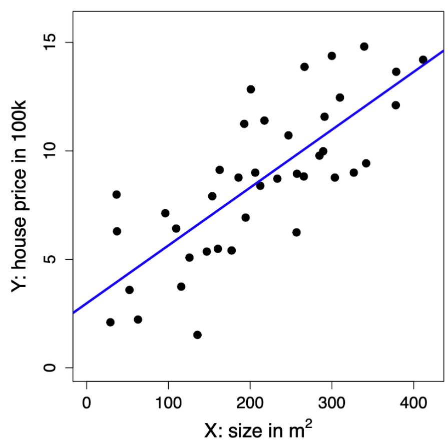


Figure 2: Regression: simple linear model

1.5 Regression: kNN method

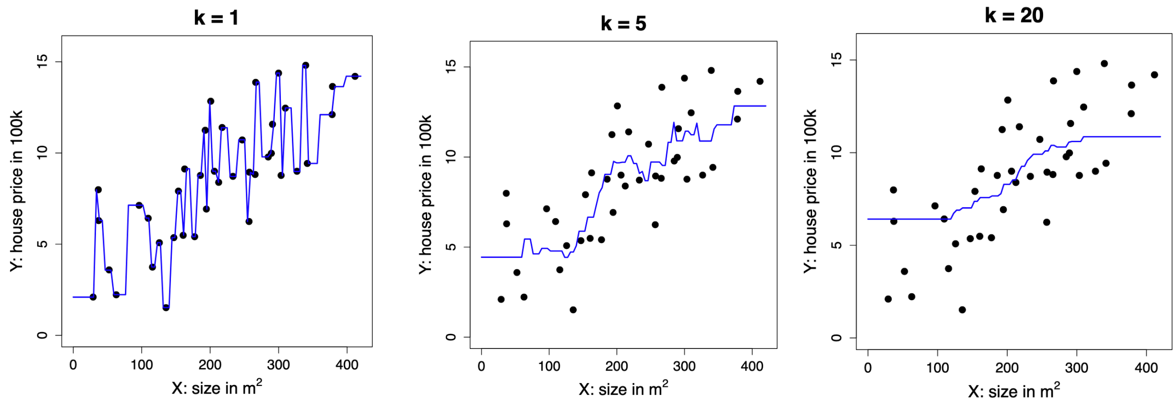
- **Data**: Observations (x1, y1), . . . , (xn, yn)
- **Goal**: Predict unknown outcome y0 for new predictor x0.

1.5.1 The k-Nearest Neighbor (kNN) Method

The prediction for a new input x_0 is given by:

$$\hat{y}_0 = \frac{1}{k} \sum_{i:x_i \in N_k(x_0)} y_i = \text{ave}\{y_i : x_i \in N_k(x_0)\}$$

- $N_k(x_0)$ refers to the k closest points x_i to x_0 in the training data.
- The number k is a **tuning parameter** that controls how many neighbours are used to make the prediction.



As we see in the figures above, the greater the number of neighbours, the smoother the curve since we are taking into account more information.

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#TODO how to put the three images next to each other with the undertitle
```

Business application: advertising

Medical application: body fat

1.6 Classification: Simple Linear Regression

- **Data:** Observations $(x_1, y_1), \dots, (x_n, y_n)$, where $y_i \in G = \{0, 1\}$

```
#TODO can classification methods have more y_i than 0 or 1?
```

- **Goal:** Predict unknown class y_0 for new predictor x_0

Directly apply linear regression treating the classes as quantitative values 0 and 1.
The prediction is

$$\hat{y}_0 = \begin{cases} 1, & \text{if } \hat{\beta}_0 + \hat{\beta}^T x_0 \geq \frac{1}{2}, \\ 0, & \text{otherwise.} \end{cases}$$

The numbers $\hat{\beta}_0, \hat{\beta}_1 \in \mathbb{R}$ are estimated model parameters.

can be explained term-by-term as follows:

- $\hat{y}_0$: The predicted class label for a new input x_0. The hat $\hat{()}$ indicates it's an estimate from the model.
- **The piecewise notation** $\{\ldots$ means the prediction depends on a condition.
- $\hat{\beta}_0$: The estimated intercept of the linear regression model, a scalar.
- $\hat{\beta}^T x_0$: The dot product (transpose of $\hat{\beta}$ times x_0), representing the weighted sum of the input features.
- $\hat{\beta}_0 + \hat{\beta}^T x_0$: The linear combination of the intercept and weighted features — the model's predicted score.
- $\geq \frac{1}{2}$: The threshold. If the score is greater than or equal to 0.5, predict class 1; otherwise, 0.
- **1 and 0**: The two possible class labels (e.g., positive/negative or class 1/class 0).

In simple terms:
You calculate a score from the model for x_0. If it's at least 0.5, you predict class 1; otherwise, class 0.

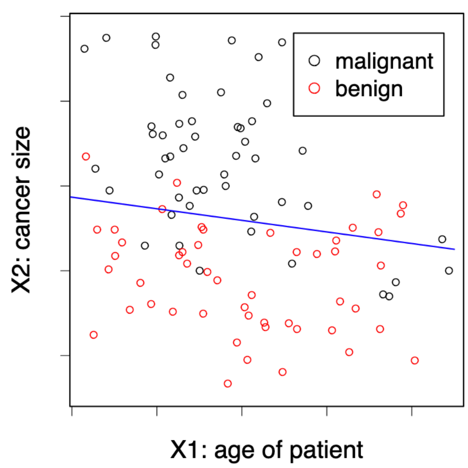


Figure 3: Classification: simple linear model

1.7 Classification: the kNN Method

- **Data:** Observations $(x_1, y_1), \ldots, (x_n, y_n)$, where $y_i \in G = \{0, 1\}$
- **Goal:** Predict unknown class y_0 for new predictor x_0

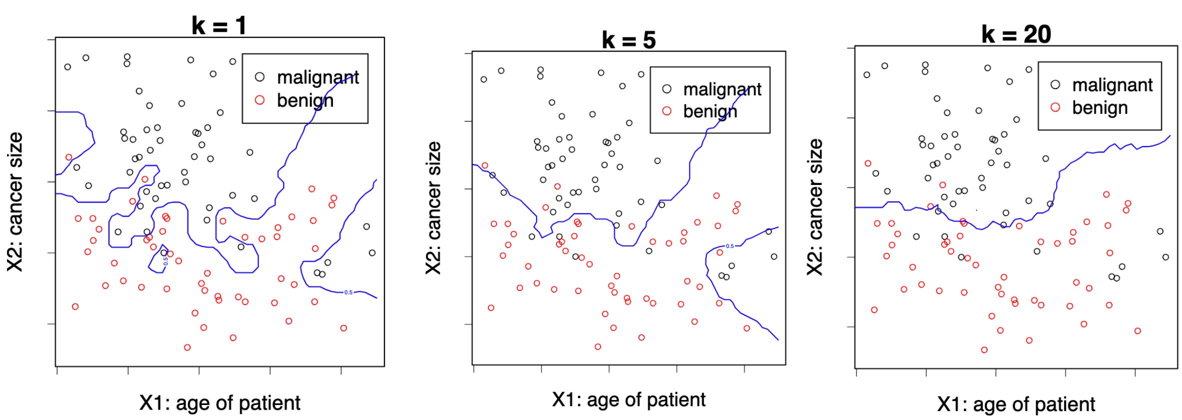
The k-Nearest-Neighbor (kNN) method predicts the majority class in the neighborhood:

$$\hat{y}_0 = \begin{cases} 1, & \text{if } \frac{1}{k} \sum_{i: x_i \in N_k(x_0)} \mathbf{1}\{y_i = 1\} > \frac{1}{2}, \\ 0, & \text{otherwise.} \end{cases}$$

Here, $N_k(x_0)$ are the k closest points x_i to x_0.
The number k is a tuning parameter, meaning that it does not come from the data. We can choose which k would be better using Cross Validation, for example.

1.7.1 Explanation of terms:

- $\hat{y}_0$: The predicted class label for the new input x_0.
- $N_k(x_0)$: The set of the k closest neighbors to x_0 in the training data.
- $\mathbf{1}\{y_i = 1\}$: An indicator function that equals 1 if the class label y_i is 1, and 0 otherwise.
- $\frac{1}{k} \sum_{i: x_i \in N_k(x_0)} \mathbf{1}\{y_i = 1\}$: The fraction of neighbours whose class is 1 — effectively, the proportion of class 1 in the neighbourhood.
- $> \frac{1}{2}$: The decision threshold — if more than half the neighbours are class 1, predict 1; otherwise, predict 0.
- k: The tuning parameter controlling how many neighbors to consider.



Again, with low k, we tend to overfit the model to our training data.

1.7.2 Handwritten characters / digits recognitions

Each pixel gives us the amount of darkness in a grey scale (0 to 9). We have 32 predictors (16 times 16, the size of the pictures). Given this predictors the system has to predict

1.8 What is the right model?

All models are wrong, some models are useful.

— George Box

In a sense its right since we would never find the model that has generated our data since Nature is too complex for this. However if they are flexible enough to capture the essence of what is happening in Nature.

Everything should be made as simple as possible. But not simpler.

— Albert Einstein

We want to create a model that represents reality but we don't want to make it too complicated, if we can make it simpler without decreasing the goodness of fit, then we need to go with the simpler. In deep learning this does not apply.

2 Regression Framework

2.1 Regression: Stochastic Framework

2.1.1 The stochastic (population) model

- We have a **quantitative response** Y and p **predictors** $X_1, \dots, X_p$.
- We assume there is a relationship between Y and $X = (X_1, \dots, X_p)$ given by

$$Y = f(X) + \varepsilon$$

  where ε is a random error term with $\mathbb{E}(\varepsilon) = 0$ (on average, errors cancel out and there is no systematic bias in the error) and $\text{Var}(\varepsilon) = \sigma^2$ (the variability of ε is constant — homoscedastic — i.e. it does not depend on X).

- The function f is **fixed but unknown**[1] and represents the **systematic information** that X provides about Y.

Example Let: $Y =$ *house price* (in 100k) and $X =$ *size* (in m^2)

Then:

$$Y = 2 + 0.03X + \varepsilon$$

which can be interpreted as: house price $= 2 + 0.03 \times$ size $+$ error

- This true relationship $f(X) = 2 + 0.03X$ is called the **population regression line**.

2.1.2 Random vectors and dimensionality

- (X, Y) is a $(p + 1)$-dimensional random vector. In this simulated example:

$$X \sim \mathcal{N}(0, \sigma_X^2), \quad Y = 2 + 0.03X + \mathcal{N}(0, \sigma_\varepsilon^2)$$

Understanding why it is $(p + 1)$-dimensional

In regression, X is not just one variable — it's a vector of predictors:

$$X = (X_1, X_2, \dots, X_p)$$

[1] It's fixed because the relationship between X and Y in the population exists — it's not random. In our housing example, there really is a true relationship between size and price. And it's unknown, we don't know what f actually is, that's what we're trying to estimate using data. When we run a regression, we build an estimate $\hat{f}(X)$ that approximates the true (but hidden) $f(X)$.

1. **What we mean by** (X, Y)

    - Each X_j represents one feature (e.g. size, number of rooms, location, etc.), and there are p such features in total.
    - The response variable Y is another random variable — the outcome we want to predict (e.g. house price).
    - So when we consider the pair (X, Y) together, we're grouping all predictors and the response into one joint random vector.

2. **Counting the dimensions**

    - X alone has p dimensions, because it contains p predictor variables.
    - Y adds one more dimension (the outcome variable). Hence, when combined:

    $$(X, Y) = (X_1, X_2, \dots, X_p, Y)$$

    - This vector has $p + 1$ components → therefore, it's a $(p + 1)$-dimensional random vector.

3. **Why it matters**

    - We call it a **random vector** because all its components (the predictors $X_1, \dots, X_p$ and the response Y) are random variables.
    - Their joint distribution $P(X_1, X_2, \dots, X_p, Y)$ describes how they vary together in the population.
    - The goal of regression is to understand the **conditional relationship** between them, i.e., how Y depends on X:
    $$\mathbb{E}[Y \mid X] = f(X)$$
    - That conditional expectation is exactly the **regression function** we try to estimate.

Example

At the **population level**, (X, Y) represents the theoretical random vector:

$$(X, Y) = (X_1, X_2, \dots, X_p, Y)$$

Each X_j is a random variable corresponding to one predictor, and Y is the random response variable.

At the **observation level**, we work with realised data points — specific numerical values drawn from this population. For observation i, we write:

$$(x_i, y_i) = (x_{i1}, x_{i2}, \dots, x_{ip}, y_i)$$

For example, if $p = 2$ (size and number of rooms) and Y is house price:

Observation	x_{i1} (size)	x_{i2} (rooms)	y_i (price)
1	120	4	6.8
2	85	3	5.1
3	150	5	8.2

For the first observation:

$$(x_1, y_1) = (x_{11}, x_{12}, y_1) = (120, 4, 6.8)$$

Thus, each (x_i, y_i) is one **realisation** of the $(p+1)$-dimensional random vector (X, Y).

2.2 Regression: The Statistical Framework

2.2.1 The Training Data

- Assume we are given n measurements $(x_1, y_1), \ldots, (x_n, y_n)$ of (X, Y), where $x_i = (x_{i1}, \ldots, x_{ip})^\top$, and
  - the inputs $x_i \in \mathbb{R}^p$ are called **predictors**
  - the outputs y_i are called **responses**
- We will use this data to **learn** the unknown relationship f between Y and X, i.e., to **train** (or **estimate**) a predictive model $\hat{f}$

Example : $[Y =$ house price (100k); $\quad X =$ size (m^2)$]$; house price $\approx \hat{f}(\text{size}) = 2.977 + 0.027 \times$ size

- The training data are realizations of the random vector (X, Y):

Observation	X (Size in m²)	Y (House Price in 100k)
(x_1, y_1)	327	9.0
(x_2, y_2)	154	7.9
(x_3, y_3)	233	8.7

2.2.2 Why do we estimate f?

1. **Prediction** : Machine Learning

- A set of inputs X are readily available, but Y cannot be easily obtained.
- We can predict Y by: $\hat{Y} = \hat{f}(X)$ where $\hat{f}$ is our **predictive model** for f.

- The function $\hat{f}$ is often treated as a **black box**: we are not concerned with the exact form of $\hat{f}$ provided it yields good prediction for Y.

2. **Inference**: Classical Multivariate Analysis

- We are often interested in understanding **how** Y is affected as $X_1, \dots, X_p$ change.
- Cannot treat $\hat{f}$ as a black box; we need to know the exact form.
- We want to answer questions such as:
  - Which predictor is associated with the response?
  - What is the relationship of the response and each predictor (positive, negative, ...)?
  - Is the relationship linear or more complicated?
  - etc.

2.3 What is a good $\hat{f}$?

2.3.1 Understanding the optimal prediction

- For a fixed value of X, say $X = 3$, we want to know what would be an **optimal prediction** $\hat{f}(X)$ for the response Y.

- Since the relationship between X and Y is stochastic,[2] there might be **many different observed values** of Y corresponding to the same $X = 3$. A **good** $\hat{f}$ is the function that **minimises the expected squared prediction error**: $\hat{f}(X) = \mathbb{E}[Y \mid X]$

- In particular, for $X = 3$: $\hat{f}(3) = \mathbb{E}[Y \mid X = 3]$. That means $\hat{f}(3)$ gives the **expected value (average)** of Y for all observations where $X = 3$.

In other words, among all possible prediction functions, the conditional expectation $\mathbb{E}[Y \mid X]$ is **optimal** in the sense that it minimises the **mean squared error (MSE)**:

$$\mathbb{E}[(Y - f(X))^2]$$

Geometrically, if you look at all points with the same $X = 3$, their Y-values vary due to noise. The best prediction $\hat{f}(3)$ is the **centre** of that distribution, i.e. the average Y value for that X.

[2] Since that relationship it involves random variation, it is not perfectly deterministic .

2.3.2 Visual interpretation

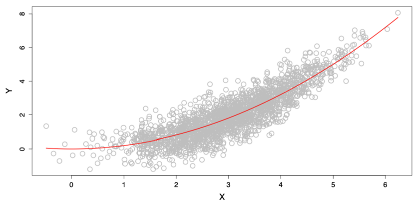


Figure 4: What is a good $\hat{f}$?

As we see in the plot on the left:

- The **red curve** represents $\hat{f}(X)$, the *conditional mean* of Y given X.
- For each value of X, $\hat{f}(X)$ passes through the *average height* of the cloud of observed points.
- This illustrates how the regression function captures the **systematic (non-random)** trend in the data, summarising the central tendency of Y as X changes.

2.4 The Loss Function

2.4.1 Defining what a "good" prediction means

To decide what makes a **good prediction**, we need a way to measure how far a predicted value $\hat{y}$ is from the true value y. This is done through a **loss function** L, which quantifies the **discrepancy** between prediction and reality.

2.4.2 Squared Error Loss

In regression, the most common choice is the **squared error loss**:

$$L(y, \hat{y}) = (y - \hat{y})^2$$

This loss penalises large errors more heavily than small ones, because the difference is squared.

2.4.3 Expected squared prediction error

For a predictive model $\hat{f}$, we can define the **expected loss**, that is, the expected squared prediction error across all possible pairs (X, Y):

$$\text{Err}_{\hat{f}} = \mathbb{E}\{[Y - \hat{f}(X)]^2\}$$

Here, the expectation $\mathbb{E}$ is taken over the joint distribution of (X, Y). It tells us the *average squared distance* between our predictions and the true outcomes in the population.

2.4.4 Finding the optimal predictor

Ideally, we would like to find a function f^* that **minimises** this expected error:

$$f^* = \arg\min_f \mathbb{E}\{[Y - f(X)]^2\}$$

This function f^* is called the **regression function**.

2.4.5 The optimal regression function

It can be shown (and we'll later prove) that the function minimising the expected squared error is:

$$f^*(x) = \mathbb{E}[Y \mid X = x]$$

That is, the **conditional expectation** of Y given $X = x$.

This satisfies the population model:

$$Y = f(X) + \varepsilon, \quad \text{where} \quad \mathbb{E}(\varepsilon) = 0.$$

- The loss function $L(y, \hat{y})$ defines how we measure mistakes.
- The expected loss $\text{Err}_{\hat{f}}$ defines how well our model performs on average.
- The best possible function f^* (the **regression function)** predicts the *mean value* of Y given X, because this minimises the expected squared error.

2.5 How do we estimate $\hat{f}$?

In theory, we know that the optimal prediction function is the **conditional expectation**:$f^*(x) = \mathbb{E}[Y \mid X = x]$. However, in practice, we often have **few or no observations** with exactly the same X value. For instance, there might be few or no data points where $X = 3$. Therefore, we **cannot compute** $\mathbb{E}[Y \mid X = 3]$ directly from the data.

What we can do to estimate $\hat{f}(x)$ in practice is to approximate the expectation ($\mathbb{E}[Y \mid X = x]$) using nearby observations to approximate. One simple and intuitive method is **k-Nearest Neighbours (k-NN) regression**.

2.5.1 The k-Nearest Neighbours approach

The **k-Nearest Neighbours (k-NN)** method estimates $\hat{f}(x_0)$ by averaging the Y values of the k data points whose X_i values are closest to x_0:

$$\hat{f}(x_0) = \text{Ave}(Y \mid X \in N_k(x_0))$$

where:

- $N_k(x_0)$ is the set of the k observations (x_i, y_i) whose x_i values are closest to x_0.
- The **number** k is a **tuning parameter**, i.e. it controls how "local" or "smooth" the estimate is.
  - If k is **small**, $\hat{f}(x_0)$ is based on very few nearby points → the estimate is **very local** and can capture detail (low bias, high variance).
  - If k is **large**, $\hat{f}(x_0)$ averages over more distant points → the estimate is **smoother** and more stable (high bias, low variance).

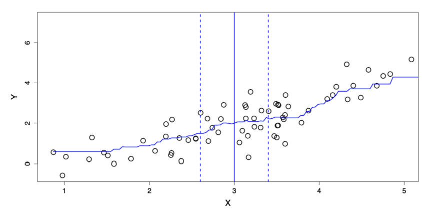


Figure 5: Estimating $\hat{f}$ with kNN

As we see in the plot on the left:

- The **vertical solid line** marks the target point $x_0 = 3$.
- The **dashed vertical lines** indicate the neighbourhood $N_k(x_0)$, which contains the k nearest observations around x_0.
- The **average of the** Y **values** within this neighbourhood provides $\hat{f}(3)$ — the *estimated regression value* at $X = 3$.

2.5.2 Limitations of k-Nearest Neighbours

Nearest neighbour averaging works well when the number of predictors (dimension) p is **small**, and the training dataset size n is **large**.

However, when the number of predictors $X_1, \dots, X_p$ increases, **kNN does not perform well**. This problem is known as the **curse of dimensionality**.

2.5.3 The curse of dimensionality

As the number of predictors p grows an data points become **sparser** in high-dimensional space. Therefore, the notion of "closeness" loses meaning since even the nearest neighbours are far away. As a result, the **averages become unreliable**, and the kNN estimate $\hat{f}(x)$ becomes noisy.

2.5.4 Towards structured methods

We will therefore study **structured methods** that:

- need **less data** to fit,
- may be more **interpretable**,
- but come at the cost of **lower flexibility**.

The key challenge is to find a good **compromise** between:

- **flexibility** (capturing complex relationships), and
- **parsimony** (simplicity and interpretability).

2.6 The empirical loss: model evaluation

2.6.1 Why the true error is unknown

The **expected prediction error** is a *theoretical quantity*:

$$\mathrm{Err}_{\hat{f}} = \mathbb{E}\{[Y - \hat{f}(X)]^2\}$$

It represents the **average squared difference** between the true Y values and the model's predictions $\hat{f}(X)$ across the entire population.

However, since we only have a finite dataset, we do **not know** the true joint distribution $P(X, Y)$. That means we cannot compute this expectation exactly, because we do not know *all possible* pairs (x, y) in the population.

Therefore, to approximate $\mathrm{Err}_{\hat{f}}$, we use a new, independent sample, called the **test dataset**:

$$\mathcal{T}_m = \{(\tilde{x}_1, \tilde{y}_1), \dots, (\tilde{x}_m, \tilde{y}_m)\}.$$

- For each input $\tilde{x}_i$, we **predict** the response using our model:

$$\hat{y}_i = \hat{f}(\tilde{x}_i)$$

- We then **compare** these predictions $\hat{y}_i$ with the observed outputs $\tilde{y}_i$.

The **empirical prediction error** (also called the **Mean Squared Error**, MSE) is:

$$\mathrm{err}_{\hat{f}} = \frac{1}{m} \sum_{i=1}^{m} [\tilde{y}_i - \hat{f}(\tilde{x}_i)]^2$$

This quantity estimates the expected squared prediction error using the available data.

We square the error for three main reasons:

1. **To penalise large errors more strongly:**
   Squaring makes large deviations more costly — an error of 4 is *four times* worse than an error of 2.
2. **To avoid cancellation of positive and negative errors:**
   Without squaring (or taking absolute values), overestimates and underestimates would cancel each other out.
3. **For mathematical and statistical convenience:**
   The squared loss is smooth and convex, making optimisation easier.
   Moreover, minimising the expected squared loss leads directly to the **regression function**:

$$f^*(X) = \mathbb{E}[Y \mid X]$$

2.6.2 Practical use

- In practice, we **split** our data into **training** and **test** sets.
- The training data are used to **fit** $\widehat{f}$, while the test data are used to **evaluate** its performance.
- When data are limited, we use **cross-validation**, which rotates the training and test roles to get a reliable estimate.
- To obtain a more reliable measure of performance, we use **cross-validation**, which consists of:
  - Dividing the dataset into K subsets (called "folds");
  - Training the model on $K - 1$ folds and testing it on the remaining one;
  - Repeating this process K times and averaging the test errors.

This ensures that every observation is used for both training and testing, giving a more robust estimate of the model's predictive accuracy.

Concept	Definition	Interpretation
$\text{Err}_{\hat{f}}$	$\mathbb{E}\{[Y - \widehat{f}(X)]^2\}$	True (theoretical) expected error
$\text{err}_{\hat{f}}$	$\frac{1}{m}\sum_{i=1}^{m}[\tilde{y}_i - \widehat{f}(\tilde{x}_i)]^2$	Empirical estimate (MSE) from test data

The squared loss penalises large errors and ensures all deviations are positive, giving a smooth, interpretable measure of how well $\widehat{f}$ generalises to unseen data.

3 Linear Regression

3.1 Simple Linear regression

Recall the stochastic model: we start from the general regression framework: $Y = f(X) + \varepsilon$ where ε is a random error term with $\mathbb{E}(\varepsilon) = 0$ and $\text{Var}(\varepsilon) = \sigma^2$.

The simple linear model

In **simple linear regression**, we assume that the relationship between Y and X is **linear**: $f(X) = \beta_0 + \beta_1 X$, where:

- β_0 is the **intercept**, i.e., the predicted value of Y when $X = 0$;
- β_1 is the **slope** — the expected change in Y for a one-unit increase in X.

Together, β_0 and β_1 are called the **model coefficients** or **parameters**.

In order to apply linear regression in practice, we first **split the data** into **training** and **test** sets. For the training data, suppose we observe n data points: $\{(x_i, y_i)\}_{i=1}^n$. We then use these observations to estimate the parameters (β_0, β_1), obtaining their fitted (sample) estimates: $(\hat{\beta}_0, \hat{\beta}_1)$ For each observation i:

- The **fitted value** (predicted response) is

$$\hat{y}_i = \hat{f}(x_i) = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

- The **residual** (prediction error) is

$$e_i = y_i - \hat{y}_i$$

Each residual measures how far the model's prediction is from the actual observed value of Y.

Finally, to measure the total prediction error across all observations, we compute the **Residual Sum of Squares (RSS):**

$$\text{RSS}(\hat{\beta}_0, \hat{\beta}_1) = e_1^2 + e_2^2 + \cdots + e_n^2$$

or equivalently,

$$\text{RSS}(\hat{\beta}_0, \hat{\beta}_1) = \sum_{i=1}^{n} (y_i - \hat{f}(x_i))^2 = \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

Minimising this RSS gives the **least squares estimates** $\hat{\beta}_0$ and $\hat{\beta}_1$ — the line that best fits the data in the least-squares sense.

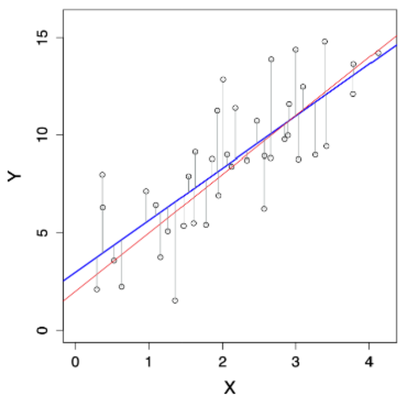


Figure 6: Simple Linear Regression: true function vs. estimated function

As we see in the plot on the left:

- The **red line** represents the *true* function $f(X) = \beta_0 + \beta_1 X$, which is generally unknown.
- The **blue line** represents the *estimated* regression line $\hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 X$.
- The **vertical grey lines** show the residuals $e_i = y_i - \hat{y}_i$, i.e., the differences between the observed and fitted values.

The goal of linear regression is to make the estimated line $\hat{f}(X)$ as close as possible to the true underlying relationship $f(X)$ by minimising the RSS.

3.2 Estimation of parameters by least squares

To estimate the coefficients (β_0, β_1) of the simple linear model, we choose the values that **minimise the Residual Sum of Squares (RSS):**

$$(\hat{\beta}_0, \hat{\beta}_1) = \arg \min_{(\beta_0, \beta_1) \in \mathbb{R}^2} \mathrm{RSS}(\beta_0, \beta_1)$$

where

$$\mathrm{RSS}(\beta_0, \beta_1) = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2$$

This means we are finding the intercept and slope that make the predicted values $\hat{y}_i = \beta_0 + \beta_1 x_i$ as close as possible to the actual y_i, in the least-squares sense.

So, **how do we solve the optimisation problem?**.

There are two main approaches to find $(\hat{\beta}_0, \hat{\beta}_1)$:

1. **Numerically**, using computer algorithms such as **gradient descent**, which iteratively adjusts the coefficients to reduce the RSS.
2. **Analytically**, by **differentiation** — setting the partial derivatives of the RSS with respect to β_0 and β_1 to zero and solving the system of equations.

The analytical (closed-form) solutions are:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

where $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ and $\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$ are the **sample means** of X and Y respectively.

Therefore, $\hat{\beta}_1$ measures the **average change in** Y associated with a one-unit increase in X, whereas $\hat{\beta}_0$ is the **predicted value of** Y when $X = 0$. Both coefficients are calculated so that the **sum of squared residuals** is as small as possible.

3.2.1 Randomness of the estimates

The estimates $(\hat{\beta}_0, \hat{\beta}_1)$ are **random variables**, not fixed numbers. This is because if we collected a new training dataset (another random sample from the same population) we would obtain slightly different estimates.

In the plot on the right:

- The **red line** represents the true model $f(X) = \beta_0 + \beta_1 X$.
- The **blue line** shows the estimated regression line $\hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 X$.
- The **grey lines** correspond to regression lines obtained from other random samples, illustrating how $(\hat{\beta}_0, \hat{\beta}_1)$ vary due to sampling variability.

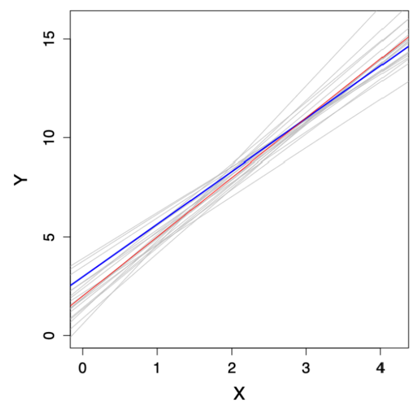


Figure 7: Sampling variability in least squares estimates

3.3 Advertising data

3.4 Multiple Linear Regression

We now consider p predictors $X = (X_1, \dots, X_p)$ for the response variable Y. The model becomes:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p + \varepsilon$$

where β_0 is the intercept (the predicted value of Y when all predictors are 0), β_j for $j = 1, \dots, p$ are the coefficients representing the partial effect of each predictor X_j, and ε is the random error term.

Intuition:
This is the natural extension of simple linear regression to multiple predictors. In simple regression, Y varies with a single X, forming a **line** in 2D space.
With multiple predictors, the model defines a **hyperplane** in $(p+1)$-dimensional space, since each predictor adds one dimension. Moreover, each coefficient β_j tells us how Y changes *on average* when X_j increases by one unit, while all other predictors are held constant.

Example: The Advertising dataset

For instance, suppose we study how different types of media advertising affect sales. Then our model could be:

$$\text{sales} = \beta_0 + \beta_1 \times \text{TV} + \beta_2 \times \text{radio} + \beta_3 \times \text{newspaper} + \varepsilon$$

Here: $Y = $ sales, $X_1 = $ TV advertising budget, $X_2 = $ radio budget, $X_3 = $ newspaper budget.

Each predictor contributes independently to explaining Y. The regression finds the "best-fitting" hyperplane that minimises the total squared distance between the observed and predicted sales values.

3.4.1 Matrix form of the training data

We observe n data points $\{(x_i, y_i)\}_{i=1}^n$, where each $x_i = (x_{i1}, \dots, x_{ip})^\top$ contains the p predictor values for observation i. All predictors can be represented compactly as an $n \times p$ matrix:

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}$$

Intuition: each **row** of **X** corresponds to one observation, and each **column** to one predictor (feature). This compact matrix notation allows us to handle all data points and predictors simultaneously, which is essential when we later express the model in vector–matrix form. In other words, x_{11} would be the value for the observation $i = 1$ coupled with the feature $x = 1$, whereas x_{32} would be the value for observation $i = 3$ coupled with the feature $x = 2$.

3.4.2 Dot product notation

For notational convenience, we sometimes write the model using a **dot product** between vectors $x, y \in \mathbb{R}^p$:

$$x^\top y = \sum_{i=1}^{p} x_i y_i = x_1 y_1 + x_2 y_2 + \cdots + x_p y_p$$

This lets us express the regression model compactly as:

$$Y = \beta_0 + x^\top \beta + \varepsilon$$

Intuition: the dot product $x^\top \beta$ is just a concise way of combining all predictors and their coefficients: $\beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p$. This compact form simplifies derivations and computations, it's the same as saying we take a weighted average of predictors, where the weights are the regression coefficients. In other words, the term $x^\top \beta$ includes all coefficients except β_0, because β_0 (the intercept) is handled separately.

3.5 Estimation and Prediction in Multiple Linear Regression

As before, we estimate the parameters $(\beta_0, \beta_1, \ldots, \beta_p)$ by minimising the **residual sum of squares (RSS)**:

$$\mathrm{RSS}(\beta_0, \ldots, \beta_p) = \sum_{i=1}^{n} (y_i - \beta_0 - x_i^\top \beta)^2$$

Expanding the dot product $x_i^\top \beta$ gives

$$\mathrm{RSS}(\beta_0, \ldots, \beta_p) = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_{i1} - \cdots - \beta_p x_{ip})^2$$

This expression measures how far the model's predictions are from the true observed values, across all n observations. The parameter estimates $(\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_p)$ are the values that **minimise** this total squared error.

Intuition: this generalises the simple linear regression idea — we're fitting a *hyperplane* rather than a line. Each coefficient β_j tells us how Y changes when predictor X_j increases by one unit, **holding all other predictors constant**. The minimisation process finds the hyperplane that best fits all the data points in p-dimensional space by minimising the vertical distances (residuals).

3.5.1 Making Predictions

Once the parameters are estimated, we can predict the response for a new observation $x_0 = (x_{01}, \dots, x_{0p})$ using

$$\hat{y}_0 = \hat{\beta}_0 + x_0^\top \hat{\beta}$$

or equivalently

$$\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_{01} + \cdots + \hat{\beta}_p x_{0p}$$

where $\hat{\beta} = (\hat{\beta}_1, \dots, \hat{\beta}_p)$.

Here, the new point x_0 represents a new combination of predictor values (e.g. TV, radio, and newspaper budgets). By plugging these into the model, we obtain the **predicted response** $\hat{y}_0$, which lies on the estimated regression surface.

In the 3D plot on the right:

- The **green–blue surface** represents the fitted regression plane (the model $\hat{f}(X)$).
- The **red points** are the observed data points (x_{i1}, x_{i2}, y_i).
- The **vertical red lines** show residuals — the distances between the actual values and the fitted surface.

Intuition: when we move from simple to multiple regression, we replace the best-fitting line with the best-fitting *plane* (or *hyperplane* in higher dimensions). The same least-squares principle applies: the plane is positioned to minimise the overall squared vertical distances between the observed points and the regression surface.

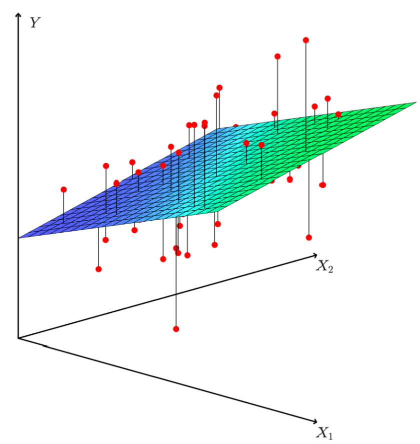


Figure 8: Fitted regression plane in multiple linear regression

3.6 Estimation and Prediction in Multiple Linear Regression

As before, we estimate the parameters $(\beta_0, \beta_1, \dots, \beta_p)$ by minimising the **residual sum of squares (RSS)**:

$$\text{RSS}(\beta_0, \dots, \beta_p) = \sum_{i=1}^{n} (y_i - \beta_0 - x_i^\top \beta)^2$$

Expanding the dot product $x_i^\top \beta$ gives

$$\text{RSS}(\beta_0, \dots, \beta_p) = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_{i1} - \cdots - \beta_p x_{ip})^2$$

This expression measures how far the model's predictions are from the true observed values, across all n observations. The parameter estimates $(\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p)$ are the values that **minimise** this total squared error.

Intuition: this generalises the simple linear regression idea — we're fitting a *hyperplane* rather than a line. Each coefficient β_j tells us how Y changes when predictor X_j increases by one unit, **holding all other predictors constant**. The minimisation process finds the hyperplane that best fits all the data points in p-dimensional space by minimising the vertical distances (residuals).

3.6.1 Making Predictions

Once the parameters are estimated, we can predict the response for a new observation $x_0 = (x_{01}, \dots, x_{0p})$ using

$$\hat{y}_0 = \hat{\beta}_0 + x_0^\top \hat{\beta}$$

or equivalently

$$\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_{01} + \dots + \hat{\beta}_p x_{0p}$$

where $\hat{\beta} = (\hat{\beta}_1, \dots, \hat{\beta}_p)$.

Intuition: the new point x_0 represents a new combination of predictor values (e.g. TV, radio, and newspaper budgets). By plugging these into the model, we obtain the **predicted response** $\hat{y}_0$, which lies on the estimated regression surface.

3.6.2 Visual Interpretation

In the 3D plot on the right: - The **green–blue surface** represents the fitted regression plane (the model $\hat{f}(X)$).
- The **red points** are the observed data points (x_{i1}, x_{i2}, y_i).
- The **vertical red lines** show residuals — the distances between actual values and the fitted surface.

Intuition: when we move from simple to multiple regression, we replace the best-fitting line with the best-fitting *plane* (or *hyperplane* in higher dimensions). The same least-squares principle applies: the plane is positioned to minimise the overall squared vertical distances between the data points and the surface.

3.7 Advertising Data

The **Advertising** dataset is a classic example of multiple linear regression, where we try to predict **sales** (in thousands of units) from the amount spent on different media types. We fit the model

$$\text{sales} = \beta_0 + \beta_1 \times \text{TV} + \beta_2 \times \text{radio} + \beta_3 \times \text{newspaper} + \varepsilon$$

Coefficient	Estimate	Std. Error	t value	p-value
Intercept	2.939	0.3119	9.42	< 2e−16
TV	0.046	0.0014	32.81	< 2e−16
radio	0.189	0.0086	21.89	< 2e−16
newspaper	−0.001	0.0059	−0.18	0.8599

The **intercept** (2.939) represents the expected sales when all advertising budgets are zero. The **TV** and **radio** coefficients are positive and highly significant ($p < 2e - 16$), meaning they have a clear effect on sales. The **newspaper** coefficient is near zero and not statistically significant ($p = 0.8599$), suggesting newspaper advertising has little to no predictive power in this model.

Intuition: each coefficient tells us how much sales are expected to change, on average, for a one-unit increase in that advertising budget, **holding other media constant**.

3.7.1 Correlations Among Predictors

	TV	radio	newspaper
TV	1	0.0548	0.0567
radio		1	0.354
newspaper			1

Intuition: correlations among predictors help us detect **multicollinearity** — when predictors are correlated with each other. Here, the correlations are low (except a moderate 0.354 between radio and newspaper), so multicollinearity is not a major issue.

3.8 Example: The Auto Data Set

The **Auto** dataset contains measurements for $n = 397$ cars, such as **mpg** (miles per gallon), **horsepower**, **year**, etc. We can fit the model

$$\text{mpg} = \beta_0 + \beta_1 \times \text{horsepower} + \beta_2 \times \text{horsepower}^2 + \varepsilon$$

Even though this model includes horsepower^2, it is still a **linear model** in the parameters $(\beta_0, \beta_1, \beta_2)$, since no coefficients are raised to a power.

Intuition: here, $X_1 = \text{horsepower}$ and $X_2 = \text{horsepower}^2$. By including the squared term, the model captures curvature since the relationship between horsepower and mpg is not strictly linear but decreases more steeply at higher horsepower levels.

In this case, both β_1 and β_2 are statistically significant, meaning the quadratic relationship is meaningful. Even higher-order polynomials (degree 3, 4, 5, etc.) can also be fitted to capture more complex patterns, but doing so increases the risk of **overfitting**.

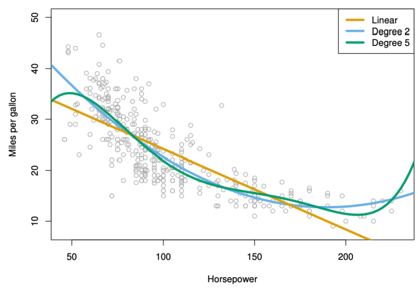


Figure 9: Different degrees Polynomial Regressions

- The **yellow line** represents a simple linear fit (degree 1).
- The **blue curve** (degree 2) captures curvature and fits the data better.
- The **green curve** (degree 5) fits the data even more closely but may overfit small variations.

The goal is to strike a balance between **bias** (oversimplification) and **variance** (overfitting) when choosing model complexity.

3.9 Beyond Linearity: Linear Basis Functions

Most often, the true function $f(X)$ is not linear in the predictors $X_1, \dots, X_p$. However, we can adjust the linear model by transforming the predictors.

Instead of using $X = (X_1, \dots, X_p)$ directly, we consider transformations of them. Let $h_m(X) : \mathbb{R}^p \to \mathbb{R}$ be the mth transformation, for $m = 1, \dots, M$. The model becomes

$$f(X) = \sum_{m=1}^{M} \beta_m h_m(X)$$

This model is still **linear in the parameters** $\beta_1, \dots, \beta_M$, but **nonlinear in the predictors** X. The functions $h_1(X), \dots, h_M(X)$ are called **basis functions**, and estimation or prediction can be applied as before using least squares.

By using basis functions, we can bend or reshape the linear model to capture more complex relationships between X and Y, i.e., without losing the simplicity of linear estimation. The model is "linear" in β but flexible in X.

3.9.1 Possible Choices for the Basis Functions $h_m(X)$

- $h_m(X) = X_m$, for $m = 1, \dots, p$: essentially recovers the original linear model.
- $h_m(X) = X_j^2$ or $h_m(X) = X_j X_k$: allows higher-order polynomial terms or interactions between predictors.

- $h_m(X) = \log(X_j)$ or $h_m(X) = \sqrt{X_j}$: captures other nonlinear transformations.
- etc.

Intuition: each $h_m(X)$ acts as a new transformed version of the original predictors, expanding the feature space to better approximate the true relationship $f(X)$. Linear regression then finds the best linear combination of these transformed predictors.

4 Gradient Descent

In least squares regression, we often minimise the residual sum of squares (RSS) to estimate the coefficients. Here, for simplicity, we fix β_1 to its true value β_1^* and only estimate β_0 by least squares.

In gradient descent, the **cost function** (often denoted as $J(\theta)$ or $L(\theta)$) is the mathematical expression that measures how wrong the model is — that is, how far the model's predictions are from the true values. It tells us how costly a particular choice of parameters θ (like β_0, β_1, ...) is.

The goal of gradient descent is to find the parameters θ that **minimise the cost function**, i.e., make the model as accurate as possible. In this case, to find the value of β_0 that minimises

$$\hat{\beta}_0 = \arg\min_{\beta_0 \in \mathbb{R}} \mathrm{RSS}(\beta_0) = \arg\min_{\beta_0 \in \mathbb{R}} J(\beta_0)$$

- $\hat{\beta}_0$: this is the **estimated intercept**. The hat indicates it's an estimate obtained from the data, not the true parameter.
- arg min: this stands for *"argument of the minimum."* It means: find the value of β_0 that minimises the expression following it. For example, if you think of a function $f(x)$, $\arg\min_x f(x)$ gives the x that produces the lowest possible value of $f(x)$.
- $\beta_0 \in \mathbb{R}$: the parameter β_0 is a real number.
- $\mathrm{RSS}(\beta_0)$: this is the **Residual Sum of Squares** as a function of β_0. It measures the total squared difference between the observed y_i and the predicted values given β_0:

  $$\mathrm{RSS}(\beta_0) = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2$$

  Since β_1 is fixed here, RSS only changes with β_0.
- $J(\beta_0)$: this is a **generic name for the cost function**. In optimisation problems, we often denote the function to be minimised as $J(\cdot)$. Here, $J(\beta_0)$ is simply another name for $\mathrm{RSS}(\beta_0)$.

4.1 The Gradient Descent Algorithm

- Start with an arbitrary initial value $\beta_0^{(0)} \in \mathbb{R}$.
- In the ith step:
  - Compute the **gradient** (or derivative) of the cost function J at the current position $\beta_0^{(i-1)}$:

$$\nabla J(\beta_0^{(i-1)}) = \frac{\partial J(\beta_0)}{\partial \beta_0}\Big|_{\beta_0=\beta_0^{(i-1)}}$$

  - Update your estimate using:

$$\beta_0^{(i)} = \beta_0^{(i-1)} - \alpha \nabla J(\beta_0^{(i-1)})$$

    where α is a **tuning parameter** known as the **learning rate**.
- Repeat until convergence, i.e. when $\beta_0^{(i)}$ changes very little or not at all.

Gradient descent is an iterative algorithm that adjusts parameters step by step in the direction that most rapidly decreases the cost function (here, RSS). The learning rate α controls how large each step is:

- If α is too small $\rightarrow$ convergence is very slow.
- If α is too large $\rightarrow$ the algorithm may overshoot and fail to converge.

This process continues until the algorithm reaches the minimum of $J(\beta_0)$, i.e., the point where its slope is zero.

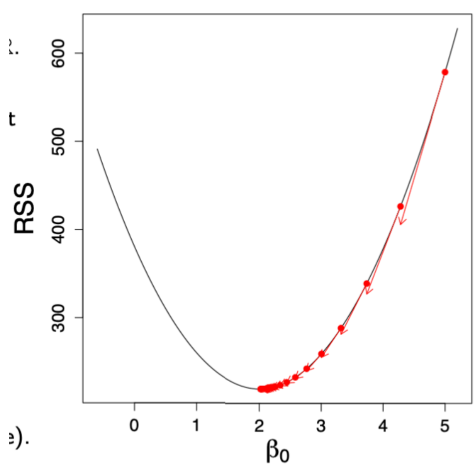


Figure 10: Gradient descent minimising the cost function $J(\beta_0)$

4.2 Gradient Descent: Higher Dimensions

In multiple linear regression, we estimate both β_0 and β_1 (and, in general, all β_j for $j = 1, \ldots, p$) by minimising the cost function:

$$(\hat{\beta}_0, \hat{\beta}_1) = \arg \min_{(\beta_0,\beta_1)\in\mathbb{R}^2} \text{RSS}(\beta_0, \beta_1) = \arg \min_{\theta\in\mathbb{R}^d} J(\theta)$$

Here $\theta = (\beta_0, \beta_1)^\top$ denotes the parameter vector, and $J(\theta)$ (or RSS) measures the total squared error between predictions and observed values.

4.3 The Gradient Descent Algorithm (in multiple dimensions)

- Start with an arbitrary initial value $\theta^{(0)} \in \mathbb{R}^d$.
- In the ith step:
  - Compute the **gradient** of the cost function J at the current position $\theta^{(i-1)}$:

$$\nabla J(\theta^{(i-1)}) = \frac{\partial J(\theta)}{\partial \theta}\Big|_{\theta=\theta^{(i-1)}} \in \mathbb{R}^d$$

  - Update your estimate using:

$$\theta^{(i)} = \theta^{(i-1)} - \alpha \nabla J(\theta^{(i-1)}) \in \mathbb{R}^d$$

    where α is the **learning rate** (a tuning parameter controlling the step size).

- Stop when the algorithm **converges**, i.e. when $\theta^{(i)}$ changes very little or not at all.

- The plot shows **contour lines** of the cost surface $J(\beta_0, \beta_1)$, where each line connects points with equal RSS.
- The **red dot** marks the minimum — the optimal pair $(\hat{\beta}_0, \hat{\beta}_1)$.
- Gradient descent iteratively updates (β_0, β_1), moving step by step along the steepest downward direction until it reaches the centre.

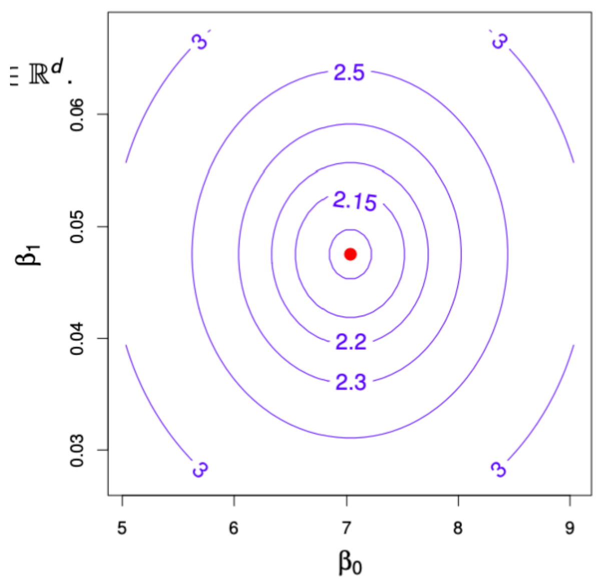


Figure 11: Gradient descent in higher dimensions: contour plot of the cost function $J(\beta_0, \beta_1)$.

4.4 Gradient Descent: Remarks

Gradient descent is a general optimisation algorithm used to find the parameter values that minimise a cost (or objective) function $J(\theta)$.nFormally, it solves the problem $\arg\min_{\theta \in \mathbb{R}^d} J(\theta)$.

4.5 Key Points

- The essential components are:
  - The **objective function** $J(\theta)$, which measures model error or loss.
  - Its **derivatives** (gradients) in all directions, denoted ∇J.
- The **gradient** $\nabla J(\theta)$ is the collection of **derivatives** of J with respect to each parameter in θ. Intuitively, a derivative measures how much the cost function J changes when one parameter changes slightly.
  In one dimension, it's just the slope $\frac{dJ}{d\theta}$; in multiple dimensions, the gradient vector
  $\nabla J(\theta) = \left[\frac{\partial J}{\partial \beta_0}, \frac{\partial J}{\partial \beta_1}, \cdots, \frac{\partial J}{\partial \beta_p}\right]^\top$
  points in the direction of the **steepest increase** of J at point θ. Therefore, gradient descent moves **in the opposite direction** — the direction of the **steepest decrease** — to minimise $J(\theta)$.
- **Important restriction:** the function J must be **differentiable**. Otherwise, gradients cannot be computed.
- Gradient descent is a **generic numerical method** that typically finds a **local minimum** of $J(\theta)$.
- If $J(\theta)$ is **convex**, then the local minimum is also the **global minimum**, and convergence is guaranteed.
- The **tuning parameter** α, also called the **learning rate**, determines how large each update step is. It must be carefully chosen or adapted. If too large, the algorithm overshoots the minimum; if too small, it converges very slowly.
- In complex or high-dimensional models, gradient descent is often the **only feasible way** to find even a local minimum.
- Depending on the model, there exist several **variants and approximations**, such as:
  - **Stochastic Gradient Descent (SGD)**: uses a subset (batch) of data per iteration.
  - **Backpropagation**: applies gradient descent efficiently in neural networks.

Think of gradient descent as walking downhill on a complex landscape. You can't always see the entire surface, but by feeling the slope beneath your feet (the gradient), you can take small steps in the direction that lowers your elevation (the loss) until you reach a valley, ideally, the lowest one.

5 Training versus Test Errors

5.0.1 Training Data

- We fit our model $\hat{f}$ on the **training data** $\{(x_1, y_1), \ldots, (x_n, y_n)\}$, representing n observations of (X, Y).
- The model learns patterns from this dataset by adjusting its parameters (e.g. β_0, β_1, …) to minimise the error between predictions and true values.
- The **training error** measures how well the model fits this known data:

$$\text{MSE}_{\text{Tr}} = \frac{1}{n}\sum_{i=1}^{n}\{y_i - \hat{f}(x_i)\}^2$$

- Because the model has already seen these points, increasing its complexity (e.g. more parameters or higher-order terms) almost always reduces this error, sometimes to zero.
- However, this makes the training error **overly optimistic**, since it does not reflect how the model performs on unseen data.

5.0.2 Test Data

- To evaluate **generalisation**, we use a separate **test dataset** $\{(\tilde{x}_1, \tilde{y}_1), \ldots, (\tilde{x}_m, \tilde{y}_m)\}$ containing m new observations not seen during training.
- The model $\hat{f}$, trained earlier, is now applied to these new points to produce predictions $\tilde{y}_i = \hat{f}(\tilde{x}_i)$.
- The **test (or generalisation) error** is then computed as:

$$\text{MSE}_{\text{Te}} = \frac{1}{m}\sum_{i=1}^{m}\{\tilde{y}_i - \hat{f}(\tilde{x}_i)\}^2$$

- This test error provides an unbiased measure of how well the model is expected to perform on future data.
- Formally, it estimates the **expected prediction error:**

$$\text{Err}_{\hat{f}} = \mathbb{E}[(Y - \hat{f}(X))^2]$$

The training error measures how well the model explains the data it has already seen, while the test error shows how well it generalises to unseen examples. As model flexibility increases, the training error steadily decreases, but the test error often follows a U-shaped curve, first improving, then worsening, because of **overfitting**.

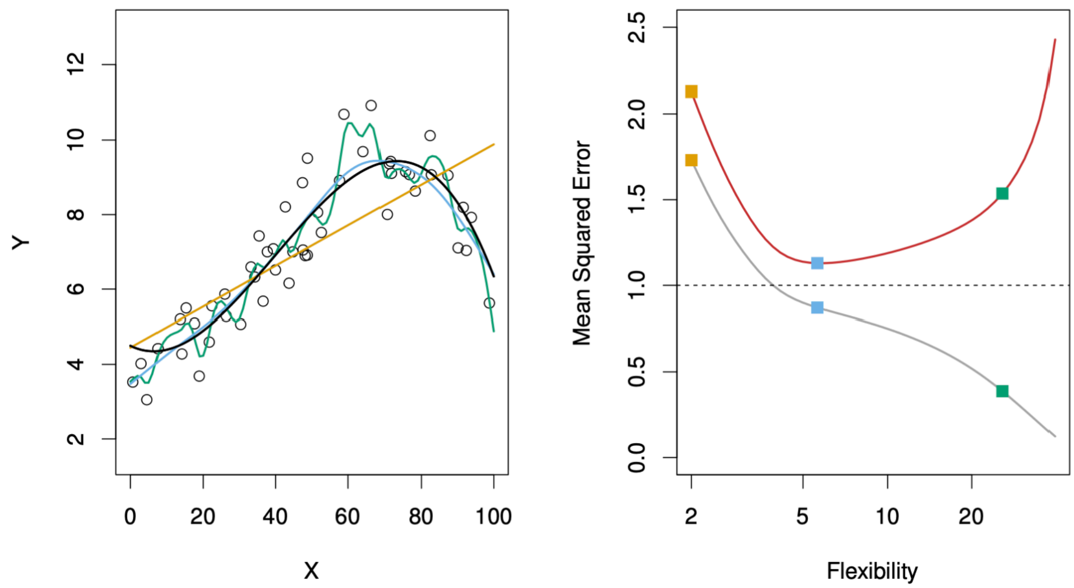


Figure 12: Training versus test errors

The figure above illustrates how **training** and **test errors** evolve as model **flexibility** increases.

- On the **left**, the data are generated from the true function f (black line) and fitted using three models:
  - a **linear model** (orange line),
  - a **moderately flexible polynomial** (blue line),
  - and a **highly flexible polynomial** (green line).
- On the **right**, the curves show:
  - **Training error** MSE_{Tr} in grey, which decreases steadily as flexibility increases.
  - **Test error** MSE_{Te} in red, which decreases at first but eventually rises due to overfitting.

When model flexibility is low, both training and test errors are high, in other words, the model underfits and fails to capture the structure of the data. As flexibility increases, the model fits the training data better and generalises more effectively, reaching a point of **optimal complexity** where test error is minimised. Beyond this point, further flexibility causes the model to memorise noise, leading to rising test error, i.e., the hallmark of **overfitting**.

5.1 The Bias–Variance Decomposition

- We assume a simple model where the outcome is generated as

$$Y = f(X) + \varepsilon, \quad \text{with } \mathbb{E}(\varepsilon) = 0 \text{ and } \text{Var}(\varepsilon) = \sigma_\varepsilon^2.$$

- For a regression fit $\hat{f}$, the **expected prediction error** at an input $X = x_0$ is

$$\text{Err}_{\hat{f}}(x_0) = \mathbb{E}[(Y - \hat{f}(X))^2 \mid X = x_0].$$

- This can be decomposed into three components:

$$\begin{aligned}\text{Err}_{\hat{f}}(x_0) &= \sigma_\varepsilon^2 + \text{Bias}^2(\hat{f}(x_0)) + \text{Var}(\hat{f}(x_0)) \\ &= \text{Irreducible Error} + \text{Bias}^2 + \text{Variance}.\end{aligned}$$

Intuition:
The **Irreducible Error** is the error due to the noise variable ε and cannot be improved. The **Bias** measures how far the model's average prediction $\mathbb{E}[\hat{f}(x_0)]$ is from the true function $f(x_0)$.
The **Variance** captures how much $\hat{f}(x_0)$ fluctuates across different training sets.
A good model finds a balance between low bias and low variance to minimise total error.

5.1.0.1 Example: Bias–Variance Trade-off in k-Nearest Neighbours**

The k-Nearest Neighbours (k-NN) algorithm predicts $\hat{f}_k(x_0)$ as the average of the k closest training points to x_0:

$$\hat{f}_k(x_0) = \frac{1}{k} \sum_{x_i \in N_k(x_0)} y_i.$$

The expected prediction error is

$$\text{Err}_{\hat{f}}(x_0) = \sigma_\varepsilon^2 + \left[f(x_0) - \frac{1}{k} \sum_{x_i \in N_k(x_0)} f(x_i)\right]^2 + \frac{\sigma_\varepsilon^2}{k}.$$

When k is small, the model is very flexible → low bias, high variance. As k increases, predictions become smoother → higher bias, lower variance. The total prediction error reaches its minimum when bias and variance are optimally balanced.

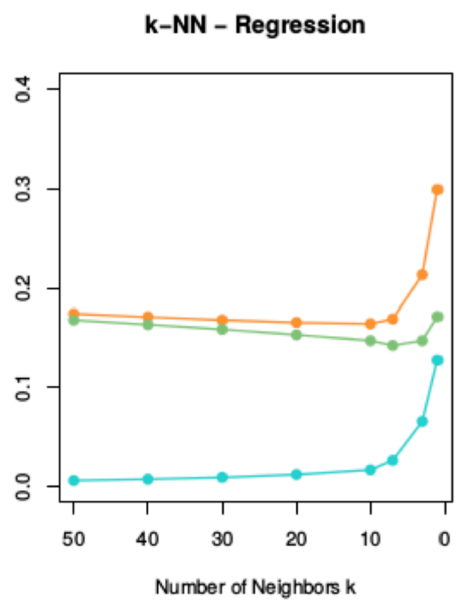


Figure 13: Bias–variance decomposition in k-NN

Intuition:
- When k is **small**, the model is very flexible → **low bias** but **high variance**.
- As k increases, predictions become smoother → **higher bias** but **lower variance**.
- The **orange curve** in the figure shows the **total prediction error**, which is minimised at an intermediate k, where bias and variance are optimally balanced.

- The **green curve** represents the **squared bias**, and the **blue curve** shows the **variance**, both plotted as functions of k to illustrate their trade-off.

Summary:
- **Low** k (high flexibility): low bias → high variance → overfitting
- **High** k (low flexibility): high bias → low variance → underfitting
- **Optimal** k: balanced bias and variance → minimal expected prediction error

5.2 The Bias–Variance Trade-off

Simple parametric models such as **linear regression** or **logistic regression**:

- Have **low flexibility** → **high bias**, but require less data to fit.
- Are **stable** across different datasets → **low variance**.
- Fit well when the data satisfy model assumptions.
- Are suitable for **inference** and **interpretation**, since model parameters have clear meaning.

Complex (often non-parametric) models such as **k-Nearest Neighbours** or **decision trees**:

- Can be **highly flexible** → **low bias**, but require more data to train.
- Are more prone to **overfitting** and can vary strongly across datasets → **high variance**.
- Perform poorly in **high-dimensional** settings (large p) due to the **curse of dimensionality**.
- Are often better suited for **prediction**, where flexibility outweighs interpretability.

In other words, simple models generalise better when data are limited or assumptions hold, while complex models can capture richer patterns but risk memorising noise.

Choosing the right model involves balancing **bias**, **variance**, and the **purpose** of analysis, i.e., explanation or prediction.

When we start with a very simple model, it has **high bias**, it systematically misses important structure in the data. Both training and test errors are large because the model is too rigid and underfits. As we make the model more flexible, it begins to capture the true relationships in the data, reducing bias and fitting the underlying pattern more accurately. Since this structure exists in both the training and test sets, the **test error initially decreases**.

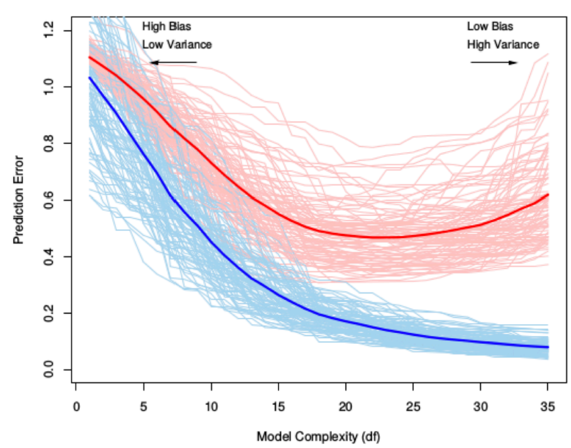


Figure 14: Training versus test error

Beyond a certain level of flexibility, however, the model becomes **too sensitive** to the training data. It starts fitting random noise or small fluctuations that are not part of the true signal. On the training data, the error keeps dropping, but these fitted noise patterns do not generalise to new observations. As a result, variance, the model's sensitivity to random noise, dominates, and the **test error rises again**.

The **U-shaped curve** of the test error reflects the sum of two competing effects:

$$\text{Test Error} = \text{Bias}^2 + \text{Variance} + \text{Irreducible Error}$$

As model complexity increases, **bias² decreases** (the model fits better) while **variance increases** (the model becomes unstable across datasets). Initially, bias reduction dominates, leading to lower test error; eventually, the growing variance outweighs this benefit, causing the test error to rise again.

6 Cross-Validation

- The **training error** MSE_{Tr}[3] is **not reliable** for model selection, since a model that overfits the training data can achieve a very small MSE_{Tr} but perform poorly on unseen data.
- A good model should minimise the **expected prediction error** $\text{Err}_{\hat{f}}$, that is, it should generalise well to new data. In other words, $\text{Err}_{\hat{f}}$ measures how well will our model predict outcomes for new data that it has never seen before.
- However, in practice, we only have **one dataset**, so we must find a way to **estimate** $\text{Err}_{\hat{f}}$ using it.

Two main approaches:

1. **Theoretical approximations** of $\text{Err}_{\hat{f}}$
   - Based on analytical criteria such as the **Akaike Information Criterion (AIC)** or the **Bayesian Information Criterion (BIC)**.
   - These rely on assumptions about the model class (e.g. linear models) and penalise excessive complexity to prevent overfitting.
2. **Empirical estimation** of $\text{Err}_{\hat{f}}$
   - Based on the **test error** MSE_{Te} estimated from resampling methods such as **Bootstrap** or **Cross-Validation**.
   - These methods approximate the model's performance on unseen data by dividing the available dataset into subsets used for training and testing.

[3]The **Mean Squared Error (MSE)** measures the average squared difference between the predicted values $\hat{y}_i$ and the true values y_i: $\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$. It quantifies how close the model's predictions are to the actual outcomes. In cross-validation, MSE is computed on the validation folds to estimate the model's **generalisation ability**. **Lower MSE values indicate better predictive performance on unseen data, making it the most common metric for model comparison and selection in machine learning**.

Intuition: Cross-validation provides a systematic way to simulate how the model would perform on new, unseen data. By repeatedly splitting the dataset into training and validation parts, we can estimate the expected prediction error $\text{Err}_{\hat{f}}$ and choose the model complexity that minimises it.

6.1 K-Fold Cross-Validation

- We only have **one dataset** $\{(x_1, y_1), \dots, (x_n, y_n)\}$. The question is: *how can we estimate how the model will perform on new, unseen data*
- In **K-Fold Cross-Validation**, we randomly divide the dataset into K equally sized subsets, called **folds**, denoted $C_1, C_2, \dots, C_K$.
- Each fold acts once as a **validation set**, while the remaining $K - 1$ folds are used for training.
- We then compute the **validation MSE** for each fold:

$$\text{MSE}_k = \frac{K}{n} \sum_{i \in C_k} (y_i - \hat{f}_{-k}(x_i))^2$$

  where $\hat{f}_{-k}$ is the model trained on all data *except* fold C_k.
- Finally, the **K-Fold Cross-Validation estimate** of the prediction error is the **average validation error** across all folds:

$$\text{CV}_{(K)} = \frac{1}{K} \sum_{k=1}^{K} \text{MSE}_k$$

Since we cannot collect new data to test our model, we simulate this process by holding out different subsets of the data in turn. Each fold provides an estimate of how well the model generalises, and averaging these results gives a more stable and less biased estimate of the **expected prediction error** $\text{Err}_{\hat{f}}$.

Choosing the number of folds:
The parameter K determines how the data are divided between training and validation, directly influencing the **bias**, **variance**, and **computational cost** of the cross-validation estimate.

- With a **small** K (e.g. 5): each **training set** is smaller because a larger portion of the data is held out for validation in each iteration. As a result, the model is fitted on less data and may **underfit slightly**, leading to a **higher bias** in the estimated prediction error. However, the **validation sets are larger**, which makes the error estimates across folds more stable and results in **lower variance**. Training fewer models also makes the procedure **computationally faster**.
- With a **large** K (e.g. 10, or $K = n$ in Leave-One-Out Cross-Validation): each **training set** is larger, and the **validation set** becomes smaller. This reduces bias since each model is trained on nearly the full dataset, but increases variance because the validation results depend more on specific observations. It also requires fitting more models, which increases **computational cost**.

In practice, choosing $K = 5$ or $K = 10$ usually provides the best balance: bias remains reasonably low, variance manageable, and computation time practical for most applications.

6.2 Interpreting the K-Fold Cross-Validation Estimate

- The **K-Fold Cross-Validation estimate** is a widely used and general method that can be applied to almost any model.

- This estimate allows us to **select the best model** among several candidates and provides an empirical approximation of the **expected prediction error** $\mathrm{Err}_{\hat{f}}$ for the final chosen model.

- The appropriate choice of K depends on the dataset size n, but in practice, $K = 5$ or $K = 10$ are commonly used as they provide a good balance between bias, variance, and computational efficiency.

Interpreting the cross-validation estimate:
K-Fold Cross-Validation not only estimates how well a model generalises, but also gives an indication of the **uncertainty** associated with this estimate, that is, how much the result would vary if we repeated the procedure with a different random split of the data. This variability is measured through the **standard error** of the cross-validation estimate:

$$\widehat{\mathrm{SE}}(\mathrm{CV}_{(K)}) = \frac{1}{\sqrt{K}} \sqrt{\frac{\sum_{k=1}^{K} (\mathrm{MSE}_k - \mathrm{CV}_{(K)})^2}{K-1}}$$

A **small standard error** means that the model performs consistently across folds. Its generalisation is stable and reliable.
A **large standard error**, on the other hand, indicates that the model's performance varies substantially depending on which data points are used for training and validation, suggesting possible instability or overfitting.

Thus, cross-validation helps assess not only *which model performs best on average*, but also *which model performs most consistently* across different subsets of the data.

6.3 Leave-One-Out Cross-Validation

- One limitation of **K-Fold Cross-Validation** is that it requires training K separate models, which can become computationally expensive for large datasets.

- A useful special case of K-Fold CV is **Leave-One-Out Cross-Validation (LOOCV)**, where each observation is treated as its own validation fold. In this case, $K = n$ and $C_k = \{k\}$ for $k = 1, \ldots, n$.

- The **LOOCV estimate** of the prediction error is:

$$\mathrm{CV}_{(n)} = \frac{1}{n}\sum_{k=1}^{n}(y_k - \hat{f}_{-k}(x_k))^2$$

  where $\hat{f}_{-k}$ denotes the model fitted on all data except the k-th observation.

- In general, this requires fitting n separate models (one for each observation) which can be computationally intensive. However, for **linear models**, a convenient approximation exists:

$$\mathrm{CV}_{(n)} \approx \frac{1}{n}\sum_{k=1}^{n}\left(\frac{y_k - \hat{f}(x_k)}{1 - S_{kk}}\right)^2$$

  where S_{kk} are leverage values (diagonal elements of the hat matrix), automatically available from the linear model fit.

- Overall, $\mathrm{CV}_{(K)}$ is a reliable estimate of the **expected prediction error** $\mathrm{Err}_{\hat{f}}$. The choice of K reflects a **bias–variance trade-off**:

  – Large K (e.g. LOOCV) → smaller bias but higher variance.
  – Small K (e.g. 5-Fold CV) → higher bias but lower variance.

In practical terms, LOOCV makes almost full use of the data for training in each iteration, which **reduces bias** because each model is trained on nearly all available observations. However, since each validation fold contains only one observation, the resulting errors vary more from case to case, leading to **higher variance**. This makes LOOCV less stable for complex models or large datasets, but highly effective and appealing for small samples or simple models such as linear regression.

6.4 Cross-validation applied to the body fat dataset

The body fat dataset is used to illustrate how **K-Fold Cross-Validation (CV)** helps in selecting the most suitable regression model when multiple predictors are available.As seen in the plot on the right:

- The **y-axis (CV)** represents the average cross-validation error.
- Each point indicates the CV estimate for a model with a given number of variables.
- The **grey bars** show one standard error around that estimate.
- The model with the **lowest CV error** (black dot) achieves the best predictive performance under this random partitioning of the data.

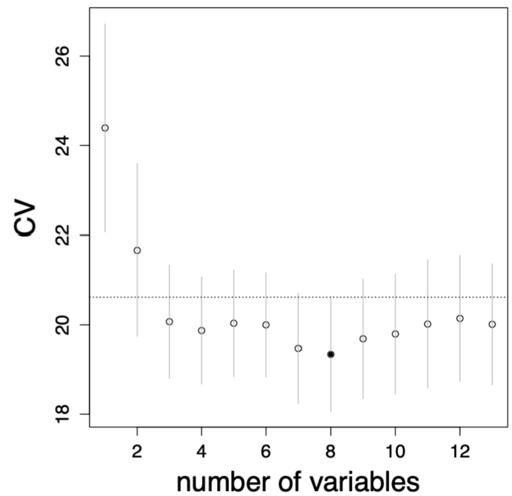


Figure 15: 10-fold CV

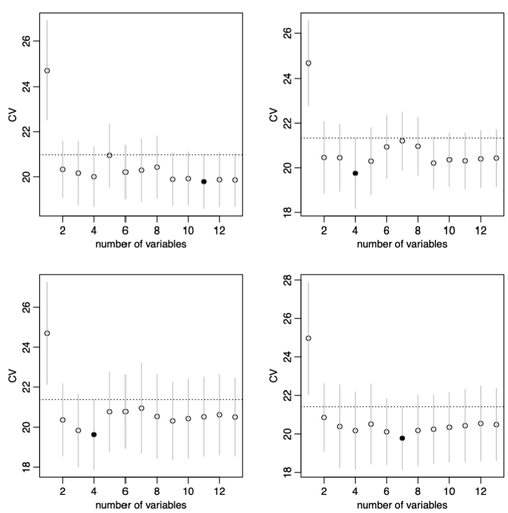


Figure 16: Four repetitions of 10-fold CV

When the folds are randomly selected:

- Repeating the **10-fold CV** several times can lead to **different results**.
- The **minimum CV error** does not always occur at the same number of predictors.
- This variation arises because **CV depends on the random partitioning** of the data.
- Therefore, a single CV run may not always yield a stable estimate of the optimal model size, in other words, **randomness matters**.

Table 6

Model sizes (k) selected across 100 repetitions of 10-fold cross-validation.

k	1	2	3	4	5	6	7	8	9	10	11	12	13
arg min of CV	0	5	8	11	0	1	11	20	11	12	18	3	0
one-sd-error rule	0	83	15	2	0	0	0	0	0	0	0	0	0

6.4.1 Stability and the one-standard-error rule

The instability caused by random fold assignments can be mitigated by applying the **one-standard-error rule**. Instead of always picking the model with the absolute lowest CV error, we choose the **simplest (most parsimonious)** model whose error is within **one standard error** of the minimum. This rule favours models that generalise better and are less sensitive to random fluctuations in the folds.

To illustrate this, the cross-validation process was repeated **100 times**. Table 6 shows how often each model size (k) was selected:

These results show that **direct minimisation** of the CV error produces more variable choices across repetitions, whereas the **one-standard-error rule** consistently selects smaller and more stable models (most frequently $k = 2$). This confirms that the rule provides a **more robust and interpretable** choice, i.e., sacrificing only a negligible loss in predictive accuracy for a significant gain in simplicity and reliability.

6.5 Final model selection

Cross-validation identified a model with **three predictors** ($\hat{k} = 3$): **weight**, **abdomen**, and **wrist**.
As shown in Figure 15, this specification achieved the **lowest cross-validation error** in the *best-subset selection* analysis, striking an effective balance between predictive accuracy and simplicity. It explains most of the variation in body fat percentage without overfitting.

The corresponding model $\hat{f}_3$ is then refitted on **all 252 observations** to obtain the final parameter estimates based on the full dataset.

Overall, cross-validation proved useful not only for identifying the best-performing model ($p = 3$) but also for ensuring that the final specification remains **interpretable, stable, and generalisable**.

7 Linear Model selection and Regularization

7.1 Credit data set and data standardisation

- The **Credit data set** from the package *ISLR* contains the response *balance* (average credit card debt) and quantitative predictors such as *age*, number of credit *cards*, *income*, and *education*.
- It also includes four qualitative predictors: *gender*, *student*, *status*, and *ethnicity*.
- Such variables often exhibit **different numeric scales** or units of measurement, which can affect the behaviour of many algorithms.
- When predictors $X_1, \ldots, X_p$ have **different scales**, the relative magnitude of each variable can distort distance-based methods (e.g. k-Nearest Neighbour) or affect regularisation penalties.
- To ensure comparability, it is good practice to **standardise the predictors** in the training data $\mathbf{x}_i$, $i = 1, \ldots, n$, for each margin $j = 1, \ldots, p$:

$$x_{ij}^{\text{scaled}} = \frac{x_{ij} - \bar{x}_j}{\hat{\sigma}_j}$$

  where $\bar{x}_j$ and $\hat{\sigma}_j$ are the sample mean and standard deviation of predictor j

- In Python, this can be done with the transformer `StandardScaler()`.
- For a new input $x_0 \in \mathbb{R}^p$, standardise as $x_{0j}^{\text{scaled}} = (x_{0j} - \bar{x}_j)/\hat{\sigma}_j$ before applying the predictive model.

Standardisation ensures that all predictors contribute equally to the model by removing scale effects. Without it, a single variable with large numerical values—such as *income* measured in thousands—can dominate the model's distance or penalty calculations, overshadowing the contribution of smaller-scale variables such as *age* or *education*.

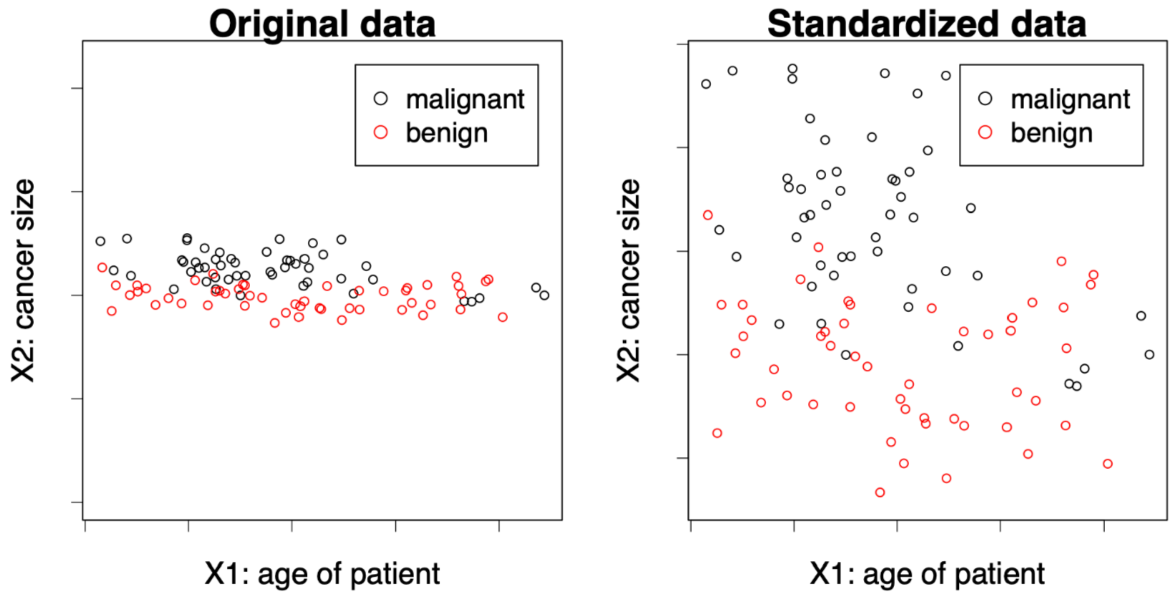


Figure 17: Effect of standardisation on feature scale

Using an unscaled variable like *income* (with values in thousands) directly in a model can lead to **biased parameter estimation or distorted decision boundaries**. The figure above shows how scale differences skew the relative importance of predictors. After standardisation, each feature is centred and rescaled to unit variance, allowing the model to treat them on an equal footing.

7.2 Back to linear models

- In the **regression setting**, a multiple linear model

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

  is fitted to the data (x_i, y_i), $i = 1, \ldots, n$.

- The estimated vector $\hat{\beta} = (\hat{\beta}_0, \ldots, \hat{\beta}_p)$ minimises the residual sum of squares (RSS):

$$\text{RSS} = \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2$$

- Why might we want to use a **different estimation method** than least squares?

(1) **Prediction accuracy:** If n is not much larger than the number of predictors p, or even $p \gg n$ (a *high-dimensional setting*), the least squares estimate $\widehat{\beta}$ may have high variance, leading to overfitting. By **shrinking** or **regularising** the coefficients, we can reduce variance and improve prediction accuracy on new test data.

(2) **Model interpretability:** Some predictors may have weak or no relationship with the response, yet least squares rarely estimates their coefficients as exactly zero. Excluding such **irrelevant predictors** simplifies the model and improves interpretability.

As we saw in Figure 8, the fitted regression plane represents the least-squares solution that minimises the vertical distances between the observed data points and the estimated surface. However, when the number of predictors grows or when predictors are correlated, the position and slope of this plane become unstable. Regularisation methods modify this fitting process by penalising large coefficient values, thereby stabilising the plane and improving both prediction accuracy and interpretability.

7.3 Alternatives to least squares

Several approaches extend or modify least squares estimation to improve prediction accuracy and interpretability when dealing with many or correlated predictors.

(1) **Subset selection:** Identify a subset of the p predictors and then fit a linear model by least squares.

- *Best subset selection* evaluates all possible combinations of predictors.
- *Stepwise selection* (forward or backward) adds or removes predictors sequentially.
  Details in [ISL], Chapter 6.

(2) **Dimension reduction:** Project the p predictors into an M-dimensional subspace ($M < p$) using techniques such as PCA, then fit a linear regression to the projected variables by least squares. Details in [ISL], Chapter 6.

(3) **Shrinkage (regularisation):** Fit a linear regression to all p predictors, but shrink the estimated coefficients towards zero relative to the least squares fit. Depending on the regularisation method (e.g. Ridge, LASSO), some coefficients may even become exactly zero. This penalisation both stabilises estimation and can perform **automatic variable selection**.

These alternative methods adjust the standard least-squares fitting process by reducing the number of predictors, transforming them into a lower-dimensional representation, or constraining their estimated effects to achieve models that generalise better and are easier to interpret.

7.4 Ridge regression

- Instead of minimising the RSS, the **ridge regression** coefficient estimates $\hat{\beta}_\lambda^R$ minimise the penalised version

$$\sum_{i=1}^{n}\left(y_i-\beta_0-\sum_{j=1}^{p}\beta_j x_{ij}\right)^2+\lambda\sum_{j=1}^{p}\beta_j^2=\mathrm{RSS}+\lambda\sum_{j=1}^{p}\beta_j^2$$

  where $\lambda \geq 0$ is the **tuning parameter**.
- The penalty term $\lambda\sum_j \beta_j^2$ is called a **shrinkage penalty** and becomes small when all $\beta_1, \dots, \beta_p$ are close to zero.
- For each $\lambda \geq 0$, ridge regression produces a different set of coefficient estimates $\hat{\beta}_\lambda^R$.
  - If $\lambda = 0$, they coincide with the least-squares estimates.
  - As $\lambda \to \infty$, they shrink towards zero.
- Choosing an appropriate λ is crucial and is typically done through **cross-validation**.

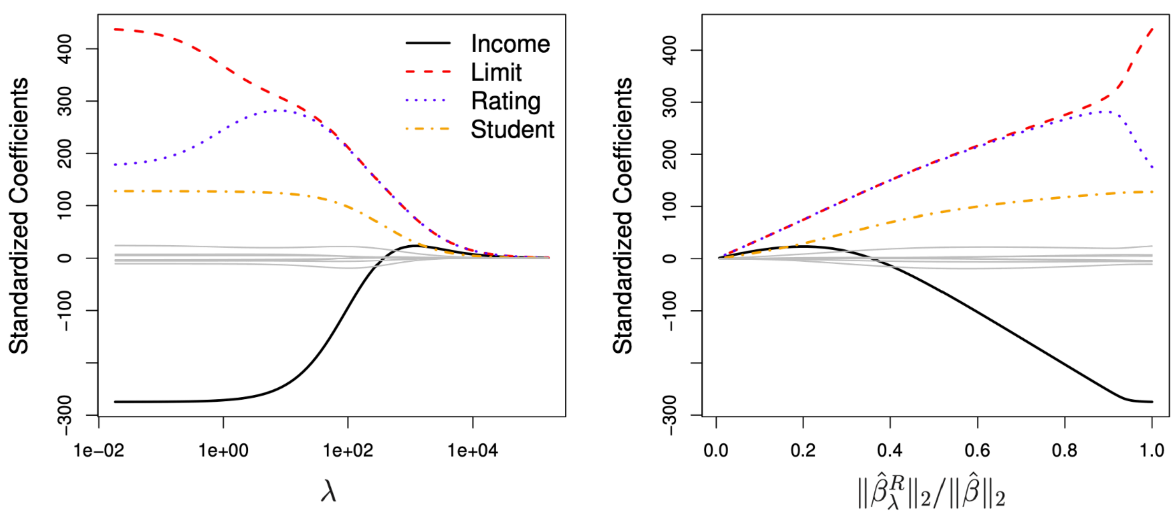


Figure 18: Ridge regression coefficient paths

As shown in Figure 18, ridge regression progressively shrinks the coefficients as the regularisation parameter λ increases.

In the **left panel**, λ is displayed on a logarithmic scale. Each coloured curve represents the standardised coefficient for one predictor. The **black curve**, corresponding to *Income*, starts around –200, indicating that higher income initially reduces the predicted credit balance. As λ increases, its magnitude gradually decreases, meaning the model gives less weight to that predictor under stronger regularisation. The

red (*Limit*), **blue** (*Rating*), and **yellow** (*Student*) curves follow similar paths, each shrinking smoothly towards zero.

In the **right panel**, the same shrinkage is shown against the ratio $\|\hat{\beta}_\lambda^R\|_2/\|\hat{\beta}\|_2$, which measures how much of the total coefficient magnitude remains relative to the unpenalised least-squares solution.

These plots illustrate the **bias–variance trade-off**: increasing λ reduces model variance but introduces bias. Ridge regression thus stabilises estimates in the presence of multicollinearity or high-dimensional data, and the optimal λ is typically identified via cross-validation.

7.5 Ridge regression: closed-form solution

- Ridge regression has a **closed-form solution**:

$$\hat{\beta}_\lambda^R = (\mathbf{X}^\mathsf{T}\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^\mathsf{T}\mathbf{y}$$

  where $\mathbf{X} \in \mathbb{R}^{n\times p}$ is the data matrix, $\mathbf{I} \in \mathbb{R}^{p\times p}$ is the identity matrix, and $\mathbf{y} = (y_1, \dots, y_n)$ is the vector of responses.

- This can be compared to the **least squares solution**:

$$\hat{\beta} = (\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}\mathbf{X}^\mathsf{T}\mathbf{y}$$

- The ridge estimator modifies the least-squares expression by adding $\lambda\mathbf{I}$ before inversion. This term prevents instability when predictors are highly correlated or when $p > n$, and shows how ridge regression **shrinks** the least squares estimates towards zero.

7.6 Lasso: least absolute shrinkage and selection operator

- Ridge regression has the disadvantage that it includes **all** p predictors in the final model.
- The **lasso** overcomes this limitation. The lasso coefficients $\hat{\beta}_\lambda^L$ minimise the penalised RSS

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}|\beta_j| = \text{RSS} + \lambda\sum_{j=1}^{p}|\beta_j|$$

  where $\lambda \geq 0$ is again a **tuning parameter**.
- The key difference from ridge regression is that the **lasso penalty** uses the ℓ_1-norm of β.
- This penalty allows some coefficient estimates to become **exactly zero** when λ is large enough, meaning the lasso performs **automatic variable selection**.
- As with ridge regression, a suitable value of λ is critical and typically chosen via **cross-validation**.

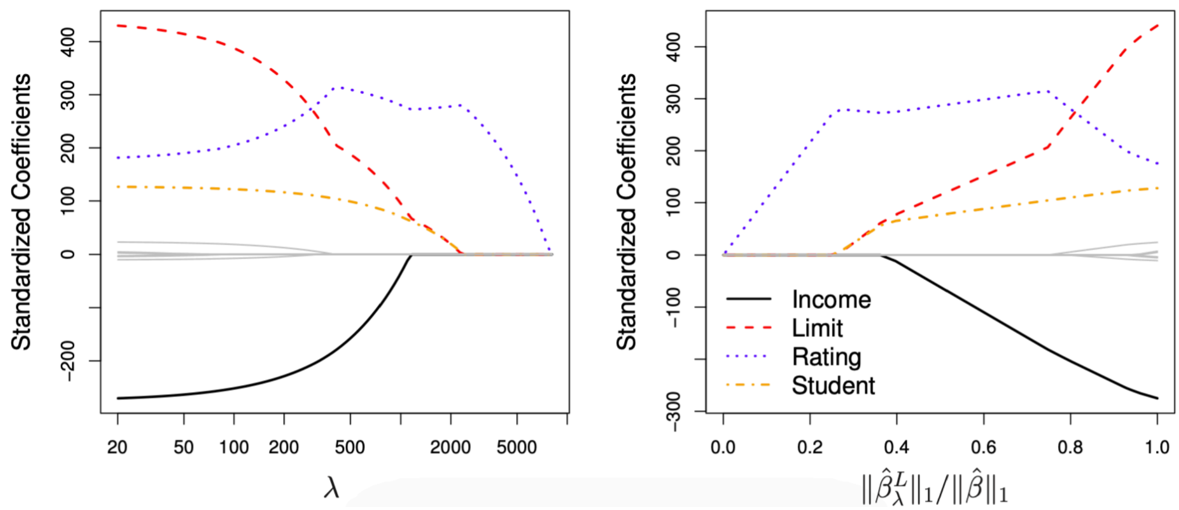


Figure 19: Lasso regression coefficient paths

As shown in Figure 19, lasso regression behaves similarly to ridge regression for small values of λ, but differs in how coefficients shrink as λ increases.
- In the **left panel**, the coefficient paths are plotted against λ on a logarithmic scale. As λ grows, some coefficients are driven exactly to zero, effectively removing those predictors from the model.
- In the **right panel**, the same process is shown against the ratio $\|\widehat{\beta}_\lambda^L\|_1/\|\widehat{\beta}\|_1$, representing the proportion of total coefficient magnitude retained.

This sparsity property distinguishes the lasso from ridge regression: while ridge continuously shrinks all coefficients, lasso can produce a simpler, more interpretable model by selecting only a subset of relevant predictors.

7.7 Regularisation: geometric interpretation

- **Equivalent formulation:** ridge and lasso regression can each be expressed as constrained optimisation problems. The estimated coefficients $\widehat{\beta}_\lambda^R$ and $\widehat{\beta}_\lambda^L$ minimise the residual sum of squares subject to different types of constraints:

$$\sum_{i=1}^{n}\left(y_i-\beta_0-\sum_{j=1}^{p}\beta_j x_{ij}\right)^2 \quad \text{subject to} \quad \sum_{j=1}^{p}\beta_j^2 \le s,$$

$$\sum_{i=1}^{n}\left(y_i-\beta_0-\sum_{j=1}^{p}\beta_j x_{ij}\right)^2 \quad \text{subject to} \quad \sum_{j=1}^{p}|\beta_j| \le s.$$

- The first corresponds to **ridge regression** (a quadratic constraint, L_2-norm), and the second to **lasso** (a diamond-shaped constraint, L_1-norm).
- This formulation provides a **geometric interpretation** of penalisation: the constraint defines a region of feasible coefficient values, and the solution occurs where this region first touches the elliptical contours of the RSS.

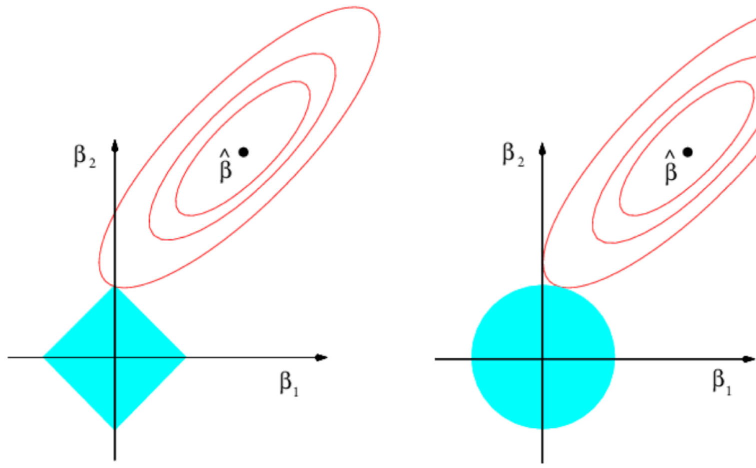


Figure 20: Geometric interpretation of lasso and ridge

As shown in Figure 20, the circular constraint region on the right represents ridge regression, where coefficients are continuously shrunk but rarely set to zero. The diamond-shaped region on the left represents the lasso: because the corners align with the coordinate axes, the optimal solution often lies exactly on an axis, producing zero-valued coefficients. This geometric property explains why the lasso can perform variable selection, while ridge regression cannot.

7.8 Comments

- For large p, best subset selection is computationally infeasible. In such cases, the **lasso** can be used to perform **variable selection** efficiently.
- While ridge coefficients $\hat{\beta}_\lambda^R$ have a closed-form solution, lasso coefficients $\hat{\beta}_\lambda^L$ do not. They are obtained through **numerical optimisation**, which remains tractable since both problems are **convex**.
- In practice, ridge and lasso often achieve similar **predictive performance** (for well-chosen λ), but lasso models tend to be **more interpretable** because of their sparsity.
- The **elastic net** (Zou & Hastie, 2005) combines ridge and lasso penalties into

$$\lambda \sum_{j=1}^{p} \left((1-\alpha)\beta_j^2 + \alpha|\beta_j| \right),$$

where $\alpha \in [0, 1]$ controls the relative weight of each penalty ($\alpha = 0$ gives ridge, $\alpha = 1$ gives lasso).

- In applied settings, lasso is often used to **pre-select** a manageable subset of relevant predictors, which can then be refined in more flexible predictive models.
- **Regularisation** extends beyond linear regression and can also be applied to models such as **logistic regression** or **generalised linear models**.

7.9 Example: Hitters data set

The **Hitters** data set from the *ISLR* package contains the annual **Salary** of 263 baseball players and $p = 19$ predictors such as the number of **Years** played in the major leagues and the number of **Hits**. The colour scale encodes salary, from low (yellow–red) to high (blue–green).
This dataset illustrates how regularisation handles models with many correlated predictors. Salary tends to increase with both *Years* and *Hits*, but these variables are also strongly correlated, leading to **multicollinearity**, an ideal case for ridge or lasso regression.

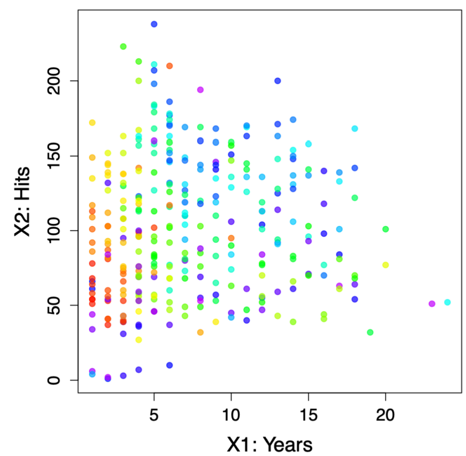


Figure 21: Hitters data set

7.9.1 Ridge regression on the Hitters data set

As shown in Figure 22, the **left panel** displays the *ridge coefficient paths* as functions of $\log(\lambda)$.
- Each coloured curve corresponds to one of the 19 predictors.
- As λ increases, coefficients shrink continuously towards zero but never reach it, meaning all variables remain in the model.

The **right panel** shows the **mean squared error (MSE)** estimated by cross-validation for different λ values.
- The dotted vertical line indicates the value of λ that minimises prediction error.
- For small λ, the model overfits; for large λ, coefficients are overly constrained and bias dominates.
- The optimal λ achieves a balance between bias and variance, providing stable but slightly biased predictions.

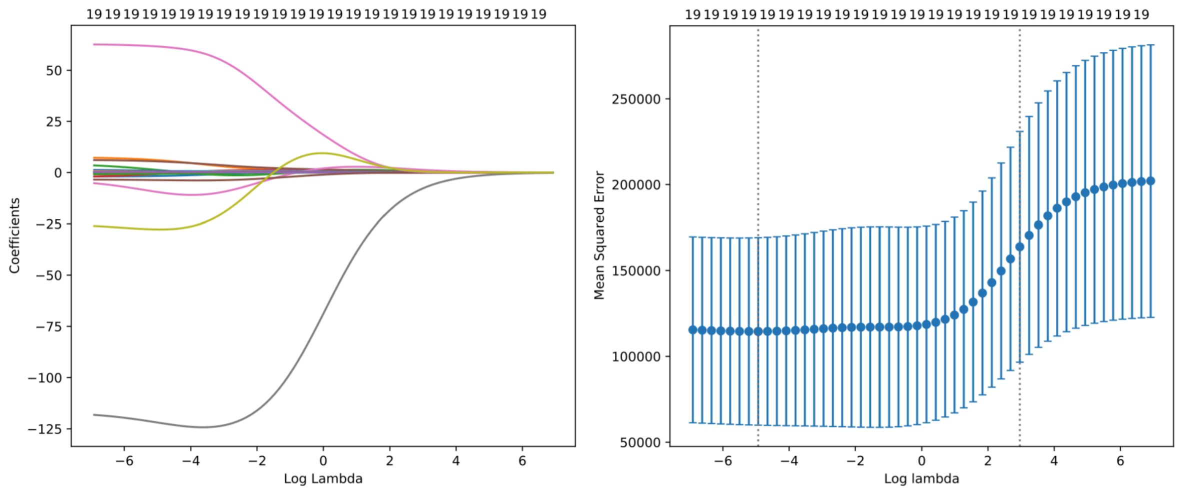


Figure 22: Ridge regression: coefficient paths and cross-validation error

7.9.2 Lasso regression on the Hitters data set

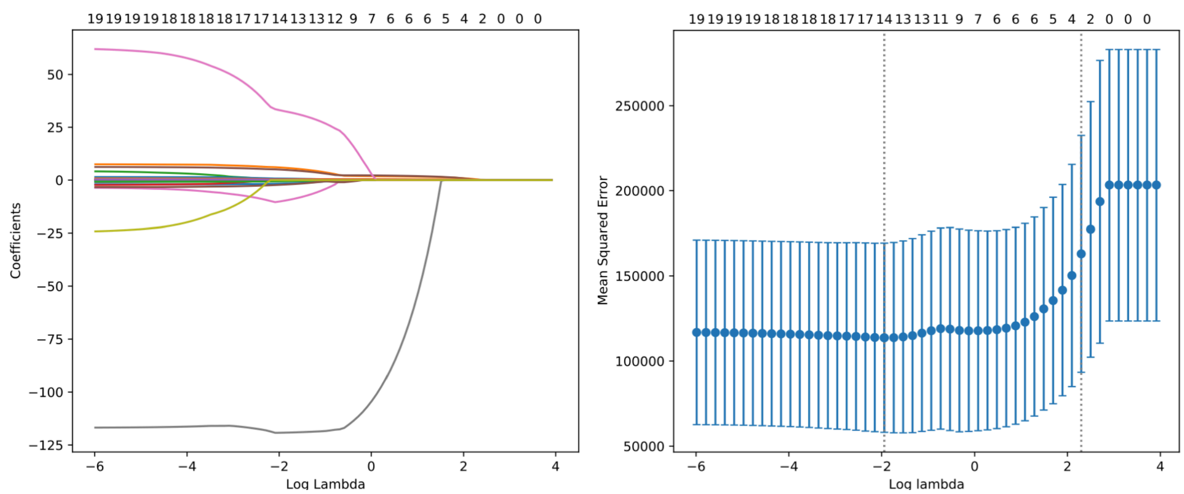


Figure 23: Lasso regression: coefficient paths and cross-validation error

In Figure 23, the **left panel** shows the *lasso coefficient paths*.
- As λ increases, some coefficients drop exactly to zero, removing the corresponding predictors from the model.
- This reflects lasso's ability to perform **automatic variable selection** by enforcing sparsity.

The **right panel** again shows the cross-validation MSE across values of λ.
- The optimal λ (dotted line) identifies a simpler model that excludes less relevant predictors while maintaining strong predictive performance.

- Compared with ridge, lasso yields more interpretable models by retaining only a subset of significant predictors.

Together, these results demonstrate how both ridge and lasso manage the bias–variance trade-off, but differ in **model complexity**: ridge smooths coefficients continuously, while lasso introduces sparsity by setting some exactly to zero.

7.10